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United States Patent	12401161
Kind Code	B2
Date of Patent	August 26, 2025
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Shield terminal and electrical connection structure including the same

Abstract

Provided are a connector and a shield terminal with which an increase in the size of the connector can be avoided. A shield terminal **30** is attached to a housing **20** from one direction that intersects a front-and-rear direction. The shield terminal **30** includes outer conductors **32** and **33** that constitute an outer shell. The outer conductors **32** and **33** include an engaging portion **43** that protrudes in the one direction from a surface thereof oriented in the one direction, and that can engage with the housing **20**.

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Appl. No.: 17/989505

Filed: November 17, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230163530 A1	May. 25, 2023

Foreign Application Priority Data

JP	2021-190347	Nov. 24, 2021
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Publication Classification

Int. Cl.: H01R13/6581 (20110101); H01R13/42 (20060101)

U.S. Cl.:

CPC **H01R13/6581** (20130101); **H01R13/42** (20130101);

Field of Classification Search

CPC: H01L (3/6581); H01L (3/42); H01L (3/73); H01R (24/38)

USPC: 439/606.01

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority from Japanese Patent Application No. 2021-190347, filed on Nov. 24, 2021, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present invention relates to a shield terminal and a connector.

BACKGROUND

(3) JP 2013-222684A and JP H5-59773U disclose a shield terminal that includes an outer conductor. The outer conductor includes a main body portion, and a barrel portion that is connected to a shield portion of an electric wire, on the rear side relative to the main body portion.

(4) Note that the outer conductor is referred to as an “outer conductor” in JP 2013-222684A, and a

“connector outer shell portion” in JP H5-59773U. The shield terminal is referred to as an “L-shaped coaxial connector” in JP 2013-222684A, and a “female connector” in JP H5-59773U. The main body portion is referred to as a “tubular portion” in JP 2013-222684A, and a “tubular coupling portion” in JP H5-59773U. The barrel portion is referred to as a “shield wire crimping portion” in JP 2013-222684A, and a “crimp portion” in JP H5-59773U.

SUMMARY

(5) Incidentally, a shield terminal is desirably provided with an engaging portion to be engaged with a housing when the shield terminal is attached to and accommodated in an accommodation groove of the housing from above. A configuration can be adopted in which, for example, a shield terminal is provided with an engaging portion having a shape protruding laterally from a barrel portion thereof, and this engaging portion is fitted into a recessed portion of a housing. However, if a plurality of shield terminals are aligned lateral to a housing, and the above configuration is adopted, the recessed portion is positioned between the accommodation grooves of adjacent shield terminals, and thus it is not possible to reduce the pitch of the accommodation grooves, giving rise to a problem in that the size of the connector will increase.

(6) In view of this, an object of the present disclosure is to provide a shield terminal and a connector with which an increase in the size of the connector can be avoided.

(7) A shield terminal according to the present disclosure is a shield terminal that is to be attached to a housing from one direction that intersects a front-and-rear direction, and includes an outer conductor that constitutes an outer shell, and the outer conductor includes an engaging portion that protrudes in the one direction from a surface thereof that is oriented in the one direction, and is disposed so as to be capable of engaging with the housing.

(8) According to the present disclosure, it is possible to provide a shield terminal with which an increase in the size of the connector can be avoided.

(9) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view showing a state where a shield terminal is accommodated in a housing, in a first embodiment of the present disclosure.

(2) FIG. 2 is a side view of the shield terminal.

(3) FIG. 3 is a perspective view of a first outer conductor.

(4) FIG. 4 is a perspective view of a second outer conductor in an open state.

(5) FIG. 5 is a plan view of a front end side of the shield terminal.

DETAILED DESCRIPTION

(6) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here.

(7) First, embodiments of the present disclosure will be listed and described.

(8) A shield terminal according to the present disclosure includes: (1) a shield terminal that is to be attached to a housing from one direction that intersects a front-and-rear direction, and includes an outer conductor that constitutes an outer shell, and the outer conductor includes an engaging portion that protrudes in the one direction from a surface thereof that is oriented in the one direction, and that is disposed so as to be capable of engaging with the housing.

- (9) Since the engaging portion has a shape protruding in one direction, which is an attachment direction of the shield terminal, from a surface thereof oriented in the one direction, it is possible to avoid an increase in the size of the connector in the width direction that intersects the attachment direction of the shield terminal.
- (10) (2) Preferably, the outer conductor includes a main body portion that is to be connected to a partner outer conductor, a barrel portion that is to be connected to an electric wire, rearward of the main body portion, and a base portion that is positioned between the barrel portion and the main body portion, and the engaging portion has a shape obtained by folding a portion of the base portion.
- (11) Usually, the barrel portion is bent so as to extend along the outer periphery of the electric wire, by performing a crimping process. Therefore, if the engaging portion is provided on the barrel portion, it is difficult to ensure positional accuracy and dimensional accuracy of the engaging portion. In this regard, with the above configuration, the engaging portion is provided at a portion of the base portion that is positioned between the barrel portion and the main body portion, and thus it is easy to ensure positional accuracy and dimensional accuracy of the engaging portion. In addition, it is also possible to increase the degree of freedom with which the engaging portion can be formed.
- (12) (3) Preferably, the main body portion has a shape extending in the one direction, and includes an opposing surface that is oriented rearward so as to oppose the engaging portion.
- (13) The engaging portion can fall within the range in which the main body portion is formed, without independently protruding in the one direction of the shield terminal, and thus, it is possible to avoid an increase in the size of the shield terminal in the one direction due to existence of the engaging portion.
- (14) (4) Preferably, the engaging portion is shaped as a plate piece, and is disposed with a plate surface thereof oriented in the front-and-rear direction, and a reinforcing portion formed by pushing a portion of the engaging portion so as to expand forward is formed at a root portion on a front surface side of the engaging portion.
- (15) When, for example, the electric wire is pulled rearward, a rear surface of the engaging portion (a surface on the rear side of the engaging portion) is pressed firmly against a corresponding surface of the housing, and there is a concern that the engaging portion will be subjected to plastic deformation. In this regard, with the above configuration, the reinforcing portion is formed at the root portion on the front surface side of the engaging portion, and thus it is possible to avoid a situation where the engaging portion is subjected to plastic deformation.
- (16) (5) In addition, preferably, in a connector that includes the shield terminal according to the above (3) and the housing, the housing includes an engagement portion to be engaged with the engaging portion, and an insertion portion into which the main body portion is inserted, forward of the engagement portion, and the insertion portion and the engagement portion are aligned with each other in the front-rear direction.
- (17) According to the above configuration, when the electric wire is pulled rearward in a state where the shield terminal is accommodated in the housing, a force resulting from the pulling can be taken on by a portion of engagement between the engagement portion and the engaging portion. Accordingly, recession of the main body portion is restrained, and it is possible to avoid a situation where the main body portion interferes with the inner surface of the insertion portion of the housing and is deformed.

DETAILED EMBODIMENTS OF PRESENT DISCLOSURE

(18) Specific examples of the present disclosure will be described below with reference to the drawings. Note that the present invention is not limited to illustrations of these, but is indicated by the claims, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

First Embodiment

(19) As shown in FIG. 1, a connector **10** according to a first embodiment of the present disclosure includes a housing **20** and a shield terminal **30**. The shield terminal **30** is accommodated in the housing **20** in a state of being attached to a terminal portion of an electric wire **90**. Note that, in the following description, the front-rear direction corresponds to the longitudinal direction of the electric wire **90**, and the side on which the electric wire **90** is routed when viewed from the shield terminal **30** is defined as a rear side. In the up-down direction, the side on which the shield terminal **30** is attached to the housing **20** is defined as the upper side. This up-down direction is not in reference to a positional relation or a direction when the connector **10** is actually mounted in a device such as an automobile. “F”, “R”, “U”, and “D” in the drawings respectively indicate “front side”, “rear side”, “upper side”, and “lower side”.

(20) Housing

(21) The housing **20** is made of a synthetic resin, and includes an accommodation groove **21** extending in the front-rear direction, and an insertion portion **22** extending downward from the front end portion of the accommodation groove **21**, as shown in FIG. 1. The accommodation groove **21** is open upward along the entire length thereof in the front-rear direction. A covering member **23** such as a cover or another housing is mounted to the housing **20** from above in order to close the upper surface of the accommodation groove **21**.

(22) A recessed engagement portion **24** is formed in the lower surface of the accommodation groove **21** of the housing **20**. The engagement portion **24** is disposed at a position close to the front end portion of the accommodation groove **21**, near the rear side of the insertion portion **22**.

(23) Electric Wire

(24) As shown in FIG. 1, the electric wire **90** is a shield wire (coaxial electric wire), and includes a core wire **91**, an insulation coating **92**, a shield portion **93**, and a sheath **94** in order from the central axis to the outer circumference thereof. At the front end portion of the electric wire **90**, the sheath **94** and the insulation coating **92** are removed, and the core wire **91** and the shield portion **93** are sequentially exposed. The shield portion **93** is a braided wire or the like, and is connected to an earth portion (not illustrated).

(25) Shield Terminal

(26) As shown in FIG. 1, the shield terminal **30** includes an inner conductor **31**, outer conductors **32** and **33** that surround the outer periphery of the inner conductor **31**, and a dielectric body **34** that is disposed between the inner conductor **31** and the outer conductors **32** and **33**. The inner conductor **31**, the outer conductors **32** and **33**, and the dielectric body **34** are all L-shaped in a side view, and correspond to the entire shape of the shield terminal **30**. The outer conductors **32** and **33** and the inner conductor **31** are made of a conductive metal, and the dielectric body **34** is made of a synthetic resin. The dielectric body **34** electrically insulates the inner conductor **31** and the outer conductors **32** and **33** from each other. A rear end portion of the portion of the inner conductor **31** that extends in the front-rear direction is crimped and electrically connected to the core wire **91** of the electric wire **90**, and a lower end portion of the portion of the inner conductor **31** that extends in the up-down direction is electrically connected to a partner inner conductor of a partner terminal (not illustrated).

(27) As shown in FIG. 2, the outer conductors **32** and **33** form an outer shell of the shield terminal **30**, and are composed of a first outer conductor **32** and a second outer conductor **33** that are attached and connected to each other.

(28) The first outer conductor **32** is formed by performing a bending process or the like on a metal plate. The first outer conductor **32** includes a barrel portion **35** and a base portion **36** that extends forward from the barrel portion **35**. As shown in FIGS. 2 and 3, the barrel portion **35** includes a bottom portion **37** that extends in the front-rear direction. The bottom portion **37** and the base portion **36** are provided continuous to each other in the front-rear direction. The barrel portion **35** includes a right-and-left pair of front barrel pieces **38** (only one front barrel piece **38** is illustrated in FIGS. 2 and 3) that rise from a front end portion of the bottom portion **37**, a right-and-left pair of

intermediate barrel pieces **39** (only one intermediate barrel piece **39** is illustrated in FIGS. **2** and **3**) that rise from a front-rear intermediate portion of the bottom portion **37**, and a right-and-left pair of rear barrel pieces **41** (only one rear barrel piece is illustrated in FIGS. **2** and **3**) that rise from a rear end portion of the bottom portion **37**.

(29) The barrel portion **35** is crimped and connected to the electric wire **90**, and is bent into a shape having a circular cross-section or a cross-section close to a circular cross-section as a whole while curving the bottom portion **37**. The diameters of a portion of the barrel portion **35** corresponding to the intermediate barrel pieces **39** is smaller than the diameters of portions corresponding to the front barrel pieces **38** and the rear barrel pieces **41**.

(30) The base portion **36** includes a plate-like portion **42** that is flat in the front-rear direction and the right-left direction, at a position lower than the bottom portion **37**. In a central portion of the plate-like portion **42**, an engaging portion **43** is formed by folding the plate-like portion **42**.

Specifically, a notch having a square U-shape is formed in a central portion of the plate-like portion **42**, a plate piece portion of the notch is folded downward from the front end thereof, whereby the engaging portion **43** is formed. The engaging portion **43** is formed in a rectangular shape, and is disposed along the up-down direction with the plate surfaces thereof being directed forward and rearward. A reinforcement portion **44** having a triangular pyramid shape is provided at a root portion of the engaging portion **43**. The reinforcement portion **44** is shaped by pushing a portion that extends from a root portion of the engaging portion **43** to a front end side of the plate-like portion **42** so as to expand forward.

(31) The first outer conductor **32** includes a pair of rising portions **45** that rise from right and left end portions of the base portion **36**. The rising portions **45** are each formed in a plate shape extending along the up-down direction. Rear ends of the rising portions **45** are respectively coupled to the front barrel pieces **38**.

(32) On the front end side of each of the rising portions **45**, a rising main body portion **46** that is rectangular in a side view is formed to be raised to be higher than the rising portions **45**. An upper end of the rising main body portion **46** serves as a fastening portion **47** that can come into contact with a fastener portion **61** of the second outer conductor **33**, which will be described later. The fastening portion **47** of the rising main body portion **46** is disposed along the front-rear direction.

(33) The second outer conductor **33** is formed by performing a bending process or the like on a metal plate. As shown in FIGS. **2** and **4**, the second outer conductor **33** includes a cylindrical main body portion **48** and an extension portion **49** shaped as a plate piece. The extension portion **49** includes a lower extension portion **51** that is continuous to the main body portion **48** at a right angle, and an upper extension portion **53** that is continuous to the main body portion **48** via hinges **52** in an openable and closable manner to an open state (state shown in FIG. **4**) and a closed state (state shown in FIGS. **1**, **2**, and **5**). In the open state, the extension portion **49** rises from the main body portion **48**, and an upper surface opening of the main body portion **48** is open. In the closed state, the extension portion **49** is disposed in the front-rear direction, and the upper surface opening of the main body portion **48** is closed by an extension base end portion **59**, which will be described later.

(34) The main body portion **48** is electrically connected to a partner outer conductor of a partner terminal (not illustrated) while being inserted into the insertion portion **22** of the housing **20**. An elastic contact portion **54** (see FIG. **2**) that can come into contact with the partner outer conductor and that is disposed in the main body portion **48** is formed on the lower end side of the main body portion **48**. The main body portion **48** includes an opposing surface **65** that is directed rearward so as to oppose the engaging portion **43**.

(35) As shown in FIG. **4**, on the upper end side of the main body portion **48**, a right-and-left pair of engaging pieces **55** that are shaped as a plate piece are formed, and an open portion **56** that is open rearward is formed between rear ends of the engaging pieces **55**. The lower extension portion **51** shaped as a plate piece that extends rearward from a lower end of the open portion **56** of the main

body portion **48**. A rear end portion of the lower extension portion **51** is formed to have a width that is smaller than that of the front end portion of the lower extension portion **51**. The upper extension portion **53** includes a plate-piece shaped main extension portion **57** that is continuous to an upper end front portion of the main body portion **48** via the hinges **52**, and a pair of sub extension portions **58** that are bent from right and left end portions of the main extension portion **57** at a right angle.

(36) As shown in FIG. 2, the main extension portion **57** extends in the front-rear direction in the closed state. The extension base end portion **59** that closes the upper surface opening of the main body portion **48** in the closed state is formed at a position of the main extension portion **57** close to the hinges **52**. In the closed state, the extension base end portion **59** is positioned on the front side of the sub extension portions **58**, and is disposed along the front-rear direction and the right-left direction. As shown in FIGS. 4 and 5, a right-and-left pair of fastener portions **61** that protrude laterally are formed on the extension base end portion **59**. The fastener portions **61** have a rectangular shape, and protrude above the open portion **56** and on the lateral sides relative to the outer periphery of the main body portion **48**, in the closed state. The lateral protruding length of the fastener portions **61** is the same as the thickness of the rising main body portions **46**, which is the same as the width of the fastening portions **47**, or larger than the thickness of the rising main body portions **46**.

(37) The sub extension portions **58** include a pair of coupling end portions **62** that protrude downward from right and left end portions of the main extension portion **57** in the closed state, and a pair of divided portions **63** that extend rearward from the coupling end portions **62**. In the closed state, a rear end portion of the main extension portion **57**, a rear end portion of the lower extension portion **51**, and the divided portions **63** form a circular shape or a shape close to a circular shape as a whole in a rear view, and are disposed along the outer periphery of the shield portion **93** of the electric wire **90**. As shown in FIG. 2, in a state where the first outer conductor **32** and the second outer conductor **33** are attached to each other, the rear end portion of the lower extension portion **51** is covered by the bottom portion **37** of the first outer conductor **32**, and rear end portions of the main extension portion **57** and the divided portions **63** are covered by the front barrel pieces **38** of the first outer conductor **32**. As a result of the main extension portion **57**, the lower extension portion **51**, and the divided portions **63** being covered by and held by the barrel portion **35** in this manner, the attached state of the first outer conductor **32** and the second outer conductor **33** is maintained.

(38) Method and Structure for Attaching Shield Terminal

(39) The second outer conductor **33** is brought into the open state (state shown in FIG. 4), and the dielectric body **34** is inserted into the main body portion **48** from above. The inner conductor **31** connected to the core wire **91** of the electric wire **90** is accommodated in the dielectric body **34** in advance. Understandably, a configuration may also be adopted in which, if the dielectric body **34** has a lid portion **64** (see FIG. 1), the dielectric body **34** is inserted into the main body portion **48**, and then the inner conductor **31** is inserted into the dielectric body **34**, and the lid portion **64** is closed.

(40) Next, the second outer conductor **33** is brought into the closed state. In the closed state, as shown in FIG. 1, the main extension portion **57** of the second outer conductor **33** is disposed so as to be capable of coming into contact with the upper surface of the dielectric body **34**. In addition, in the closed state, as shown in FIG. 2, a gap G1 extending along the up-down direction is formed between a rear end of each engaging piece **55** and a front end of the coupling end portion **62**, and a gap G2 extending along the front-rear direction is formed between each sub extension portion **58** and the lower extension portion **51**.

(41) Next, the first outer conductor **32** is attached to the second outer conductor **33** from below. The bottom portion **37** and the base portion **36** of the first outer conductor **32** are disposed along the lower surface of the lower extension portion **51** of the second outer conductor **33**. In this state,

crimping tools (an anvil and a crimper) are applied to the barrel portion 35, and a crimping process is performed. The front barrel pieces 38, the intermediate barrel pieces 39, and the rear barrel pieces 41 are bent, and crimped and connected to the terminal portion of the electric wire 90. Specifically, the front barrel pieces 38 sandwich a rear end portion of the extension portion 49 therebetween, and are electrically connected to the shield portion 93 of the electric wire 90. The intermediate barrel pieces 39 are directly crimped and electrically connected to the shield portion 93 of the electric wire 90. The rear barrel pieces 41 are mechanically connected to the sheath 94 of the electric wire 90. Note that, in the first embodiment, crimping processes of the front barrel pieces 38, the intermediate barrel pieces 39, and the rear barrel pieces 41 may be performed at the same time or may be performed separately at time intervals.

(42) In a state where the first outer conductor 32 and the second outer conductor 33 are attached to each other, the rising portions 45 of the first outer conductor 32 overlap the outer surface side of the extension portion 49, as shown in FIG. 2, and close the gaps G1 and G2 formed in the second outer conductor 33.

(43) Moreover, as shown in FIGS. 2 and 5, the fastener portions 61 of the second outer conductor 33 oppose the fastening portions 47 of the rising main body portions 46 so as to come into contact with the fastening portions 47 in the front-rear direction. As shown in FIG. 2, the engaging portion 43 is disposed rearward of the main body portion 48 and opposes the main body portion 48 with a space therebetween.

(44) Structure for Accommodating Shield Terminal in Housing

(45) When the shield terminal 30 is accommodated in the housing 20, the main body portion 48 is inserted and fitted into the insertion portion 22 of the housing 20, and the plate-like portion 42 of the first outer conductor 32 is disposed along the lower surface of the accommodation groove 21 of the housing 20, as shown in FIG. 1. The engaging portion 43 protrudes downward from the plate-like portion 42 of the first outer conductor 32, and thus, as a result of the shield terminal 30 being lowered in an attachment process, the engaging portion 43 can enter the engagement portion 24 of the housing 20. Once the engaging portion 43 has entered the engagement portion 24, the rear surface of the engaging portion 43 (a surface on the rear side of the engaging portion 43) opposes the inner surface of the engagement portion 24 so as to come into contact with the inner surface of the engagement portion 24 in the up-down direction.

(46) When a force F (see FIG. 1) of pulling out the shield terminal 30 rearward acts due to, for example, the electric wire 90 being pulled rearward in a state where the shield terminal 30 is accommodated in the housing 20, the rear surface of the engaging portion 43 comes into firm contact with the inner surface of the engagement portion 24, and takes on the force F. Accordingly, recession of the shield terminal 30 is restrained. In addition, recession of the main body portion 48 is also restrained, and thus it is possible to avoid a situation where the opposing surface 65 of the main body portion 48 is pressed against the inner surface of the insertion portion 22 of the housing 20 and is deformed. The reinforcement portion 44 is formed on the engaging portion 43, and thus, even if the rear surface of the engaging portion 43 comes into firm contact with the inner surface of the engagement portion 24, it is possible to avoid a situation where the engaging portion 43 is subjected to plastic deformation.

(47) Incidentally, when the force F acts on the first outer conductor 32, there is a concern that the first outer conductor 32 will be inclined in a direction (upward direction UR, see FIGS. 1 and 2) of rotation about a portion where the engaging portion 43 and the engagement portion 24 are engaged with each other (a portion where the rear surface of the engaging portion 43 comes into contact with the inner surface of the engagement portion 24), and the first outer conductor 32 will be displaced relative to the second outer conductor 33 in the upward direction UR. However, in the first embodiment, the fastener portions 61 of the first outer conductor 32 come into contact with the fastening portions 47 of the second outer conductor 33 in the displacement direction, namely, the upward direction UR, and thus it is possible to prevent displacement of the first outer conductor 32

relative to the second outer conductor **33** in the upward direction UR. As a result, it is possible to maintain a state where the first outer conductor **32** and the second outer conductor **33** are properly attached to each other.

(48) As described above, according to the first embodiment, the following effects can be exerted.

(49) The engaging portion **43** is formed so as to protrude downward from the lower surface of the base portion **36** of the first outer conductor **32**, and thus it is possible to avoid an increase in the size of the shield terminal **30** in the width direction. In addition, in the housing **20**, the engagement portion **24** that receives the engaging portion **43** is formed in the lower surface of the accommodation groove **21**, and falls within the width of the accommodation groove **21**. Therefore, it is also possible to avoid an increase in the size of the connector **10** in the width direction.

(50) Moreover, the engaging portion **43** is formed by cutting and raising a portion of the base portion **36** positioned between the barrel portion **35** and the main body portion **48**, and thus it is easy to ensure positional accuracy and dimensional accuracy of the engaging portion **43**, and, furthermore, it is also possible to increase the degree of freedom with which the engaging portion **43** can be formed.

(51) Moreover, both the main body portion **48** and the engaging portion **43** are formed so as to extend in the up-down direction, and the opposing surface **65** of the main body portion **48** opposes the engaging portion **43**, and thus the engaging portion **43** does not protrude independently below the shield terminal **30**. Therefore, the size of the shield terminal **30** in the up-down direction does not increase due to the presence of the engaging portion **43**. In addition, the rib-shaped reinforcement portion **44** is formed at the root portion of the engaging portion **43** on the front surface side of the engaging portion **43**, and thus it is possible to avoid a situation where the engaging portion **43** is subjected to plastic deformation.

(52) Furthermore, according to the first embodiment, the following effects can be exerted.

(53) When, due to, for example, the electric wire **90** being pulled rearward, the first outer conductor **32** is about to become displaced relative to the second outer conductor **33** about the upward direction UR with respect to the portion where the engaging portion **43** and the engagement portion **24** are engaged with each other, the fastening portions **47** of the rising main body portions **46** of the first outer conductor **32** can come into contact with the fastener portions **61** of the second outer conductor **33**, and thus it is possible to prevent displacement of the second outer conductor **33** relative to the first outer conductor **32**.

(54) Moreover, when the electric wire **90** is pulled rearward, the force F resulting from the pulling can be taken on by the portion of engagement between the engagement portion **24** and the engaging portion **43**. Accordingly, recession of the main body portion **48** is restrained, and it is possible to avoid a situation where the main body portion **48** interferes with the inner surface of the insertion portion **22** of the housing **20**, and is deformed.

(55) Moreover, the fastening portions **47** are formed on the upper ends of the rising main body portions **46**, and thus there is no need to perform a special process for forming the fastening portions **47** on the first outer conductor **32**. In addition, as a result of the rising portions **45** closing a gap in the first outer conductor **32**, it is possible to prevent leakage and intrusion of noise.

OTHER EMBODIMENTS OF PRESENT DISCLOSURE

(56) The first embodiment disclosed here is to be construed as illustrative and not limiting in all respects.

(57) In the first embodiment, the engaging portion is formed at the base portion of the first outer conductor. However, in other embodiments, the engaging portion may also be formed in a portion of the first outer conductor other than the base portion, such as the barrel portion, as long as the engaging portion is shaped so as not to protrude outward in the width direction of the first outer conductor.

(58) In the first embodiment, the engaging portion is formed by cutting and raising a portion of the base portion. However, in other embodiments, the engaging portion may be formed by folding a

protruding portion of the base portion.

(59) In the first embodiment, the main body portion is formed so as to extend in the up-down direction. However, in another embodiment, the main body portion may also be formed so as to extend in the front-rear direction continuous to the extension portion, the base portion, the bottom portion, and the like.

(60) In the first embodiment, the outer conductor is constituted by the first outer conductor and the second outer conductor that are formed separately. However, in another embodiment, the outer conductor may also be a single member in which all of the constituent elements are integrated.

(61) From the foregoing, it will be appreciated that various exemplary embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various exemplary embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A shield terminal that is to be attached to a housing from one direction that intersects a front-and-rear direction, the shield terminal comprising an outer conductor that extends coaxially with an electric wire and includes: a rising main body portion that is to be connected to a partner outer conductor, the outer conductor and the partner outer conductor constituting an outer shell of the shield terminal, a barrel portion that is disposed rearward of the rising main body portion and is configured to be connected to the electric wire, and a flat surface that is disposed forward of the barrel portion and is oriented to face in the one direction, wherein the rising main body portion, the barrel portion, and the flat surface are integrally formed by bending a metal plate, and the outer conductor further includes an engaging portion that protrudes from the flat surface in the one direction and is disposed so as to be capable of engaging with the housing.
 2. The shield terminal according to claim 1, wherein the outer conductor further includes a base portion that is positioned between the barrel portion and the rising main body portion, and the engaging portion has a shape obtained by folding a portion of the base portion.
 3. The shield terminal according to claim 1, wherein the partner outer conductor includes a cylindrical main body portion that has a shape extending in the one direction, and the cylindrical main body portion includes an opposing surface that is oriented to face rearward so as to oppose the engaging portion.
 4. A connector that includes the shield terminal according to claim 3 and the housing, wherein the housing includes an engagement portion to be engaged with the engaging portion, and an insertion portion into which the cylindrical main body portion is inserted, forward of the engagement portion, and the insertion portion and the engagement portion are aligned with each other in the front-rear direction.
 5. The shield terminal according to claim 1, wherein the engaging portion is shaped as a plate piece, and is disposed with a plate surface thereof oriented in the front-and-rear direction, and a reinforcing portion formed by pushing a portion of the engaging portion so as to expand forward is formed at a root portion on a front surface side of the engaging portion.
 6. The shield terminal according to claim 1, wherein the engaging portion is shaped as a plate piece having a plate surface, and the plate surface of the engaging portion is oriented to face in the front-and-rear direction.
 7. The shield terminal according to claim 1, wherein the outer conductor further includes a reinforcement portion formed by pushing a portion of the engaging portion so as to expand forward, and the reinforcement portion is disposed at a root portion on a front surface side of the engaging portion.
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